

Homework 1: Foundations of Time Series — Written Report

Ahmad Avar Naggayev · June 29, 2026 · Companion to: DSCI506_HW1_Ahmad_Naggayev.ipynb

Part A — Week 1: Time-Series Components

A3 — First Contact with the Data

Trend: Clear upward linear trend from ~\$200K (Jan 2015) to ~\$390K (late 2024), approximately \$1.6K/month gain over the 10-year window.

Seasonal period: Repeating oscillations every 12 months (period = 12). Peaks appear in the Jun–Jul window; troughs in the Sep–Nov window.

Additive vs. multiplicative: The seasonal swings visually *grow* as the series level rises — early years show smaller oscillations than later years. This is the hallmark of multiplicative structure, where seasonal amplitude scales with the trend level. Confirmed quantitatively in D2 (mean–std correlation ≈ 0.98 across years).

Most obvious data-quality problem: A spike to 9,999.9 in December 2021 — more than 20× the surrounding values — is an unmistakable data-entry error visible immediately in the raw plot.

A4 — Components in the Wild (AI-Assisted)

Prompt used:

```
"Help me load three real time series from different domains – finance, weather, and economics – using Python. Use freely available sources or built-in datasets (e.g. AirPassengers, CO2, unemployment). For each, plot it and label which components (trend/seasonal/cyclic/noise) are present."
```

AirPassengers (Transport): Strong upward trend (post-WWII air travel growth) + clear 12-month seasonality (summer peaks) + multiplicative structure (swings grow with level). Minimal residual noise.

CO2/Mauna Loa (Environmental): Dominant upward trend (fossil fuel emissions) + annual seasonal oscillation (Northern Hemisphere photosynthesis cycle). Almost no cyclic component.

US Unemployment (Economics): No consistent seasonality at quarterly frequency. Clear cyclic behavior — recession spikes at ~1970, 1980, 1990, 2001, 2009 with irregular 5–10 year periods. Mild downward trend post-1982.

What I had to correct:

(1) The assistant used `statsmodels.datasets.co2` but did not set `freq='W-SUN'` after loading, causing the decomposition to error. Fixed by resampling to monthly frequency with `.resample('MS').mean()`. (2) It attempted to load `AirPassengers` via a URL that no longer exists; I switched to `sm.datasets.get_rdataset('AirPassengers')`. (3) It called the unemployment recession spikes 'seasonal' — they are cyclic because their timing and length are not calendar-fixed. Corrected the characterization in my write-up.

Part B — Week 2: Data Processing & Graphical Representation

B1 — Cleaning a Messy Series

Outlier rule: IQR×3 fence (upper bound = $Q3 + 3 \times IQR = 740.1$). The 9,999.9 value at 2021-12 exceeds this by 13×. A fixed threshold like '>1000' is fragile: as the trend pushes normal values toward 900–1000, a fixed cutoff starts flagging real observations.

Best imputation method: linear interpolation. The series has an upward trend and monthly seasonality, so gaps should be filled by bridging smoothly between flanking known values. Forward-fill is a poor choice here: it carries the last observed value

forward unchanged, systematically understating the imputed value in a trending series. For the 2019-11/12 consecutive gap, forward-fill repeats October twice — ignoring that the series was continuing its seasonal swing — distorting both trend and autocorrelation estimates.

B2 — Visualization Suite

1 Line plot: Clear upward trend from ~\$200K to ~\$390K over 10 years with consistent 12-month oscillations. The outlier at 2021-12 (9,999.9) is instantly visible.

2 Month-wise box plot: June sits highest; the Sep–Nov window is the trough. Box height grows across all months as the trend rises, consistent with multiplicative seasonality.

3 MoM histogram: Roughly symmetric around ~0.8%/month (reflecting trend), spread ± 15 –20%. The approximately normal distribution confirms the outlier was cleanly removed.

4 Lag-1 scatter: Very tight positive linear relationship — strong positive autocorrelation. Knowing last month's sales predicts this month's almost exactly. This implies AR-type models are appropriate; naive models without memory are not.

Why graphical representation matters: The lag-1 scatter reveals near-perfect consecutive autocorrelation — a fact completely invisible in a table of 120 numbers that has direct implications for model selection.

B4 — EDA Helper, Audited (AI-Assisted)

Prompt used:

```
"Write a reusable Python function profile_timeseries(df, date_col, value_col) that returns a tidy summary (date range, inferred frequency, count of missing values, count of flagged outliers using IQR×3 fence, basic stats) and a 2×2 panel of diagnostic plots: line plot, histogram of changes, lag-1 scatter, and ACF plot."
```

Three bugs found and fixed:

Bug 1 — absolute change instead of % change: The AI used `diff()` in the histogram, giving absolute dollar changes ($\sim \pm 50$) instead of percentage changes ($\sim \pm 10\%$). Caught by checking the x-axis scale. Fixed: replaced with `pct_change()`.

Bug 2 — implicit missing timestamps not detected: `isna().sum()` was called before reindexing to the full date range, so timestamps that were absent from the CSV entirely returned 0. Fixed: reindex to `pd.date_range(min, max, freq=inferred_freq)` first.

Bug 3 — wrong IQR multiplier: The AI used the standard $1.5 \times \text{IQR}$ fence. The assignment specifies $3 \times \text{IQR}$ for extreme outliers. With $1.5 \times$, borderline seasonal values would be over-flagged. Fixed: changed multiplier to 3.

Part C — Week 3: Smoothing & Moving Averages

C4 — Holt-Winters, Verified (AI-Assisted)

Prompt used:

```
"Use statsmodels ExponentialSmoothing to fit a Holt-Winters triple exponential smoothing model to store_sales training data (monthly, 108 observations). Use seasonal_periods=12 and additive or multiplicative based on the data structure. Forecast 12 months ahead. Show the code and plot the forecast."
```

AI version: The assistant defaulted to additive seasonal mode (`seasonal='add'`) without examining the data.

My independent verification: Based on D2 findings (mean–std correlation = 0.98, log transform stabilizes amplitude), I configured multiplicative seasonal mode (`seasonal='mul'`). The multiplicative model produced lower test-set RMSE, with the divergence largest at seasonal peaks — multiplicative scaling correctly amplifies the peak height as the trend rises, while additive applies a fixed seasonal increment that undershoots the later peaks.

Likely cause of discrepancy: The two models initialize seasonal components differently — multiplicative HW uses seasonal ratios while additive uses differences. This leads to structurally different extrapolation paths, compounding over the 12-month forecast horizon. The AI did not check whether seasonality was additive or multiplicative before choosing a model — a critical

verification step it skipped.

Part D — Week 4: Time-Series Decomposition

D2 — Additive or Multiplicative?

Decision: MULTIPLICATIVE. Evidence 1: Strong positive correlation ($r \approx 0.98$) between annual mean and annual standard deviation — as the trend rises, seasonal swings grow proportionally. Evidence 2: Taking $\log(\text{sales})$ reduces this correlation toward zero, confirming the seasonal component scales multiplicatively with the level. A model that assumes constant-sized seasonal swings (additive) would systematically understate peaks and overstate troughs in the later years of the series.

D3 — Decomposition: Scratch vs. Library

Seasonal indices (from-scratch multiplicative, period=12): Jan=1.126, Feb=1.179, Mar=1.143, Apr=1.121, May=1.155, Jun=1.199, Jul=1.116, Aug=0.940, Sep=0.732, Oct=0.633, Nov=0.721, Dec=0.936. Peak: June (+19.9% above average). Trough: October (−36.7% below average).

Classical vs. STL structural difference: Classical decomposition assumes a *fixed* seasonal pattern — the same 12 indices apply across all 10 years. STL (Seasonal-Trend decomposition using Loess) estimates *time-varying* seasonality that evolves over the sample, is robust to outliers via iterative re-weighting, and extends cleanly to endpoints. Classical decomposition loses 6 observations at each end due to the 2×12 moving average; STL does not.

D4 — Naive Forecast Assumptions

This approach assumes (1) the trend will continue to be exactly linear at the historical slope of ~\$1.6K/month, (2) the seasonal pattern will remain fixed — same shape and same relative magnitudes each year, and (3) the residuals are white noise with no exploitable structure. It will fail whenever any of these break: if growth accelerates or decelerates (structural break), if a shock changes the seasonal pattern, or if the residuals carry autocorrelated information that a model like ARIMA could capture explicitly. This motivates ARIMA and ETS: they model residual autocorrelation structure rather than treating it as pure noise, yielding better short-horizon forecasts and proper prediction intervals.

D5 — The Manager's One-Pager (AI-Assisted)

Prompt used:

"Given these 12 monthly seasonal indices for a retail store (Jan=1.126, Feb=1.179, ..., Dec=0.936) and an upward linear trend of approximately \$1.6K per month, write a short, non-technical explanation of the store's seasonality and growth for a manager who has never heard the word 'decomposition'."

AI draft (key claim, verbatim):

"Sales follow a predictable seasonal rhythm: summer months (July–August) consistently sell about 25% more than the average month, while winter months (January–February) typically sell about 20% less. This pattern repeats reliably every year."

Most important correction:

The AI mis-identified the peak and trough months and used rounded percentages that understated the actual trough severity. The actual seasonal peak is **June (+19.9%)**, not July–August. The actual trough is **October (−36.7%)**, not January–February which are actually above average (Jan=+12.6%, Feb=+17.9%). More critically, the AI stated the pattern is fixed in dollar terms — it is not. Because the series is multiplicative, a June 2015 seasonal premium of ~\$50K has grown to ~\$90K in 2024. Any staffing or inventory plan must account for this: the dollar impact of seasonality is not the same in 2024 as it was in 2015.

Reflection

Of the techniques covered in these four weeks, I trust **classical decomposition** most for understanding structure — when you can see the trend, seasonal, and residual components side by side, there is nowhere for the model to hide. I trust **simple**

forward-fill imputation the least: it is the easiest method to reach for but the most likely to silently corrupt a subsequent analysis by introducing artificial flat stretches that distort autocorrelation estimates. I was surprised by how much information the lag-1 scatter plot reveals instantly — the strong autocorrelation in this series is hidden in a table of 120 numbers but takes two seconds to read in a plot. The most important lesson from the AI-assisted problems is that AI code that *runs* is not the same as code that is *correct* — the B4 function produced output for every input but made a silent error (absolute change vs. percentage change) that would have propagated through every downstream interpretation without a careful audit.